1st Sit COURSEWORK Question Paper **Spring Semester 2023**

Module Code:	**CS4051NI/CC4059NI**
Module Title:	**Fundamentals of Computing**
Module Leader:	**Hrishav Tandukar** (Islington College)

Coursework Type:	**Individual**
Coursework Weight:	This coursework accounts for **60%** of your total module grades.
Submission Date:	**Friday, 12 May 2023 (Week 27)**
When Coursework is given out:	**SUNDAY, 19 March 2023 (Week 20)**
Submission Instructions:	Submit the following to Islington College's MST Platform before the due date: ● **Soft copy of the report** ● **Zip file with source code of the program**
Warning:	London Metropolitan University and Islington College takes Plagiarism seriously. Offenders will be dealt with sternly.

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PLAGIARISM

You are reminded that there exist regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understand these extracts:

Extracts from University *Regulations on Cheating, Plagiarism and Collusion*

Section 2.3: *"The following broad types of offence can be identified and are provided as indicative examples ….*

(i) Cheating: including taking unauthorised material into an examination; consulting unauthorised material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorised calculator during the examination or storing unauthorised material in the memory of a programmable calculator which is taken into the examination; copying coursework.

(ii) Falsifying data in experimental results.

(iii) Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.

(iv) Bribery or attempted bribery of a person thought to have some influence on the candidate's assessment.

(v) Collusion to present joint work as the work solely of one individual.

(vi) Plagiarism, where the work or ideas of another are presented as the candidate's own.

(vii) Other conduct calculated to secure an advantage on assessment.

(viii) Assisting in any of the above.

Some notes on what this means for students:

1. Copying another student's work is an offence, whether from a copy on paper or from a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.

2. Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. "$e = mc^2$ (Einstein 1905)". A *reference* section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with an attribution.

School of Computing, FLSC

This module is assessed by coursework (60%). For the coursework, the students are required to develop an application based on detailed guidance on given specifications. Through the coursework students should be able to:

- ✓ Develop a ***system for laptop shop***
- ✓ Describe the program
- ✓ Test the program with some sample data to demonstrate its behavior
- ✓ Write a report to present the work

- Students are required to submit weekly assignment along with related research evidence

- The guideline for the Documentation/Development part is given below:

    - Components to be included are: cover page, and table of contents, figures and tables, footer.

    - The report must have an **Introduction** section with definite goals and objectives (approx. 400 words or more).

    - The report must have a **Discussion and Analysis** section in which students need to explain how the program was developed.

1. Scenario

A laptop shop buys laptops/computers from manufacturers and sells it to various customers which may be individuals or companies. Orders for computers can be placed for the manufacturers. Likewise, customers can place orders for computers to our laptop shop (distributor).

The laptop rental shop manages information about the available computers in a text file. A program that can read the text file to display all the laptops available and make changes to the text file according to the nature of the transaction (ordering from manufacturer/selling to customer) needs to be developed. With each order or sale made, a note/receipt should be generated with the details of the transaction. The stock of a particular laptop should also be updated in the main text file itself. For example, if the shop currently has 19 pieces of Razer Blade, and a customer buys 2 of it, the stock should be updated to 17. In case the rental shop purchases 5 Razer Blade laptops, the stock should then be updated from 17 to 22 i.e. the stock should be increased by 5.

A sample format of the text file containing the information about the laptop is as follows:

Razer Blade, Razer, $2000, 20, i7 7th Gen, GTX 3060

XPS, Dell, $1976, 15, i5 9th Gen, GTX 3070

Alienware, Alienware, $1978, 24, i5 9th Gen, GTX 3070

Swift 7, Acer, $900, 12, i5 9th Gen, GTX 3070

Macbook Pro 16, Apple, $3500, 10, i5 9th Gen, GTX 3070

**1st column contains the name of the laptop, 2nd column contains name of brand, 3rd column contains the price of laptop, 4th column contains the quantity available, 5th column contains the processor details, 6th column contains details regarding the graphic card.*

Note: You can use your own format and add other information too.

A note/invoice should be generated for each transaction. When a laptop is sold to a customer, a note/invoice should be generated (as a .txt file) which must contain the name of the laptop, name of brand, name of customer, date and time of purchase, total amount without the shipping cost, the shipping cost itself and the total amount to be paid for the laptops which should include the shipping cost.

When laptops are to be ordered, a note/invoice should be generated again which should include the name of the distributor (company), name of the laptop, name of brand, date and time of purchase, net amount (total amount without VAT), VAT amount applicable i.e. 13% of the total amount, and the gross amount(total amount with VAT).

** The format of the notes/invoices is up to you. But each file should have a unique name.*

2. Algorithm

- An algorithm should be developed for the application where everything the program does should be taken into account. The algorithm should be described in steps, pseudocode and flowcharts should also be included.

3. Data Structures

- The programming should be done using data structures and operations in Python for input/output, character and string processing, and data storage.
- It can use any primitive or complex data structures which might be necessary for holding the data (pairs, lists, strings, dictionaries, etc.)
- The choice of data structures must be specified in the report.

4. Program

- The program must work in a loop, displaying the available laptops and waiting for the administrator to enter details of the customers. The program should not close unless the administrator decides to do so.
- The program must check the input data, displaying error messages whenever unwanted data is entered, for example if some string value is entered where a numerical value is expected.
- The program must be implemented in a modular way with separate functions for the different operations such as input/output, reading files, generating invoices/notes, etc.

5. Description

- The program must be described in terms of its structure and behavior.
- It can be presented using text and structural charts, flowcharts or other diagrams as needed.
    - The report must have a **Testing** section with evidence (through appropriate screen dumps) that has been carried out for the program.
    - The report must include a **Conclusion** section where they need to present their findings of the development and research (approx. 300 words).
- Demonstration of the project.
    - If any individual student is not able to justify his/her project, then the project will be kept under plagiarism.

NOTE: The technicality of the project will be judged during the demo and marked accordingly.

Marking Breakdown for Coursework	
Criteria	**Total Marks**
1. Introduction	*5*
2. Algorithm	*5*
3. Pseudocode	*5*
4. Flowchart	*5*
5. Data Structures	*5*
6. Program	
a. Correctness	*10*
b. Implementation (modularity, use of functions/classes)	*10*
c. Programing Style	*10*
d. Exception Handling	*10*
e. User interface/program usability	*10*
7. Testing	*5*
8. Conclusion	*5*
9. VIVA	*10*
10. Report Structure and Formatting	*5*
Total:	**100**